Wenli Yu

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EDUCATION

Northeastern University **May 2028**

Candidate for Bachelor of Science in Mechanical Engineering, Minor in CS

Related Coursework: Dynamics, Fluid Mechanics, Probability and Statistics, Mechanics of Materials, Materials Science, Statics, Robot Dynamics and Control, Fundamentals of Computer Science I-II, Discrete Structures, Physics I-II, Calculus I-III, Differential Equations and Linear Algebra

Activities: ASME, SWE, NU Robotics

EXPERIENCE

CPI Electron Device Business Beverly, MA

Mechanical Engineering Co-op **Jan. 2026– Jun. 2026**

- Interpreted mechanical drawings and GD&T to recreate and update outdated CAD models in SolidWorks, and initiated 60+ ECOs to capture undocumented floor changes — reducing build errors and bringing process documentation into compliance.
- Designed 50+ assembly, brazing, and testing fixtures that streamlined the build process and reduced assembly time, improving repeatability and throughput on the line.
- Performed tolerance stack-up analysis using GD&T on assemblies and components to ensure repeatable fit-up and leak-tight braze joints across production builds.
- Mechanically screened thousands of diodes that produced ~40% yield; provided failure data that drove a design change, raising yield to 100%.

NASA L'SPACE Mission Concept Academy

Mechanical Engineer and Program Analyst **Sept. 2025 – Dec. 2025**

- Collaborated with a 20-member multidisciplinary team, serving on subteams of 10 (engineering) and 5 (programmatic) to develop mission concepts for a Mars Caves Exploration Project.
- Authored 3 key system requirement documents (MCR, SRR, MDR) and contributed to a successful Preliminary Design Review (PDR); earned 8 verified NASA proficiency badges in Siemens NX and project management.
- Designed and selected materials for subsystems including chassis, suspension, mobility, robotic arms, and deployment mechanisms, ensuring structural integrity, manufacturability, and seamless system integration.
- Tracked mission cost, schedule, and scope changes, developed budget and resource estimates, and adhered to NASA's change-control procedures.

PROJECTS

SpotiPi | *Raspberry Pi, SolidWorks, Python, Spotify API* **Sept. 2026 - Present**

- Develop a custom Spotify-integrated MP3 player using a Raspberry Pi, leveraging the Spotify API for seamless music streaming and playback control.
- Engineer an intuitive user interface in Python, enabling smooth navigation and interaction with Spotify's features.
- Design an ergonomic, 3D-printed enclosure in SolidWorks, optimizing durability, portability, and user comfort.

BatHat | *Arduino, Raspberry Pi, SolidWorks* **Jan. 2025 - April 2025**

- Designed and prototyped a wearable device that uses echolocation principles to help visually impaired users navigate.
- Engineered a real-time obstacle detection system utilizing LiDAR sensors and an Arduino microcontroller, delivering intuitive alerts through haptic vibration feedback.
- Modeled an ergonomic housing in SolidWorks for comfortable daily wear and developed a supplemental program on a Raspberry Pi with a Pi Cam to execute API calls for user location.

TECHNICAL SKILLS/ INTERESTS

Hardware: Soldering, SLA 3D printing, Hand tools
Software: Java, Dr.Racket, Arduino, C/C++, JavaScript, MATLAB, HTML/CSS, Python
Computer-Aided Design: SolidWorks, Inventor, Siemens NX
Soft Skills: Project Management, Technical Communication, Team Collaboration, Time Management
Interests: Milk, Climbing, Cooking, Hay Day, Long walks

CERTIFICATIONS

JSTD-001 J Class 3 Soldering Skills Certification
SOLIDWORKS CAD Design Associate (CSWA)
Autodesk Inventor Certified User